raven

PLANT KNOWLEDGE HUB NOW

One trusted knowledge layer across your plant documents.

Raven helps operations, safety, and maintenance teams find the current source, resolve conflicting references, and assemble equipment or procedure context into cited, reviewable outputs.

PLANT QUESTION

What happens if reactor temperature exceeds design temperature?

AI ANSWER | SOURCES | EQUIPMENT

ANSWER SUMMARY

4 cited sources

- Design temperature exceedance risks equipment damage and must be avoided. [1]
- A-102-B transition cone limit at TI-1397 is 795 C. [1]
- HH trip TT/TAHH-1396 cuts burners on coil outlet overtemperature. [2]
- LL synthesis-gas flow FAL-L-1257A trips fuel gas to protect coils. [3]
- Allowable maximum temperature falls as pressure increases; stay within pressure-temperature limits. [4]

OPERATING NOTE

Stay within pressure-temperature limits and verify the current source before action.

Current revision verified

Current sources

A-102-B Reformer Operating Manual.pdf Coil outlet HH temperature protection	Pg 42
Ammonia Unit SOP - Start-up Heater.docx Low synthesis-gas flow trip logic	Pg 18
Reformer Mechanical Design Basis.pdf Transition cone design temperature limits	Pg 11

Equipment context

EQUIPMENT
A-102-B

AREA
Primary Reformer

KEY TAGS
TI-1397, TT/TAHH-1396, FAL-L-1257A

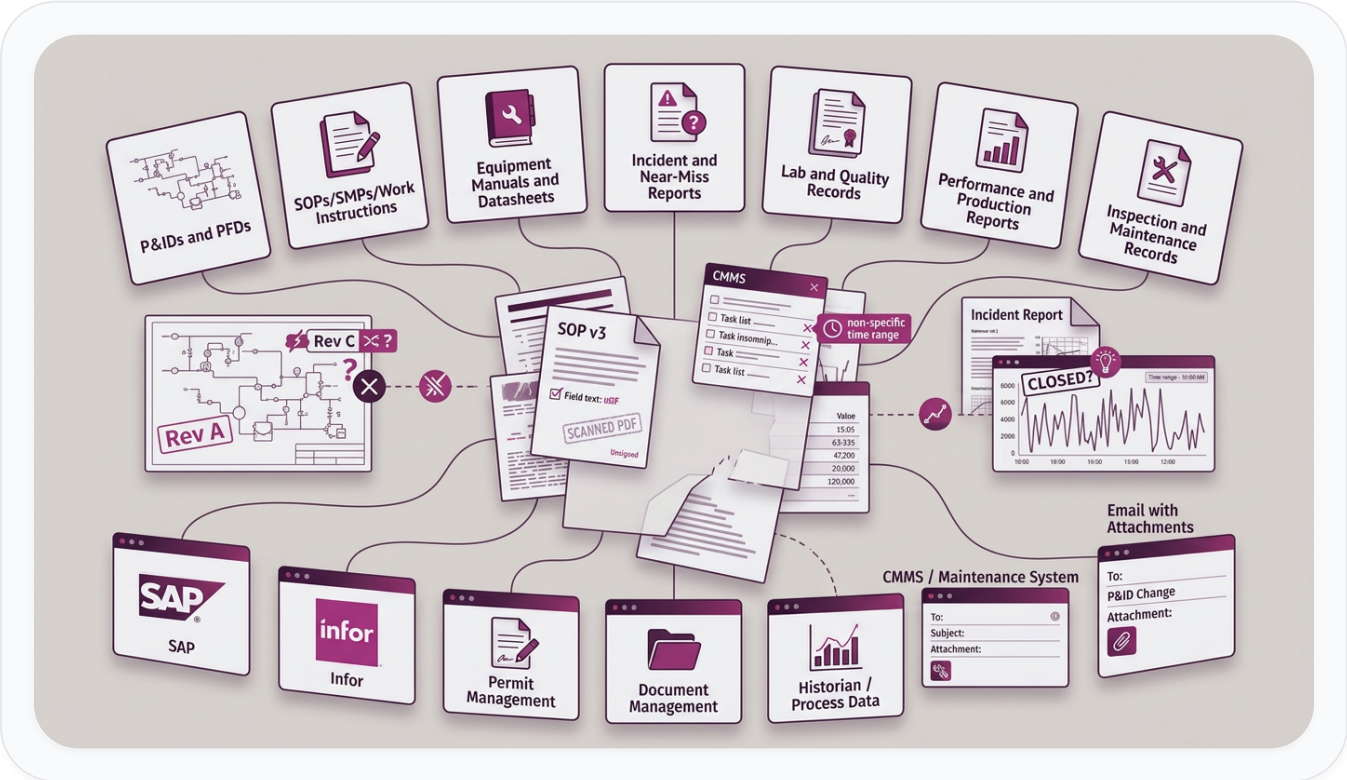
MODE
Normal operation

Is this relevant for your plant?

- Teams search across SharePoint, PDFs, email, and CMMS records before they can answer a simple plant question
- Multiple SOP or vendor-document versions make it hard to tell which reference is current
- Shift changes, leave, and workforce rotation still push people to ask the experienced person
- Startup, troubleshooting, permit, and maintenance decisions need context from more than one document

THE RISK TODAY

Before a unit restart, the shift supervisor needs the current startup procedure, the equipment limits from vendor documentation, and confirmation that the latest maintenance sign-off is complete. Each answer sits in a different place. One SOP revision is outdated but still searchable. The vendor manual is a scanned PDF. The maintenance sign-off is buried in an attachment. The information exists, but assembling the right plant context fast enough is still manual.



How Raven changes this

1. **Start from the question or the equipment**

   Ask in plain language or open a tag, procedure, or system to start from the exact plant context.

2. **Pull the right sources together**

   Raven searches across SOPs, manuals, P&IDs, incident history, and maintenance records, then surfaces the current references that matter.

3. **Resolve conflicts before the field does**

   If two documents disagree, Raven shows both values with citations so the discrepancy is visible and reviewable before someone acts on it.

4. **Turn search into a reviewable output**

   The team gets cited context they can verify, share, and use for pre-job review, troubleshooting, startup, or training.

What you get

- Cited answers with source document, page, and passage
- Cross-document discrepancy checks
- Reviewable outputs for startup, troubleshooting, and pre-job preparation
- Current-document and revision awareness
- Equipment and procedure context pulled into one view
- Audit trail for what was asked, returned, and verified

REVIEWABLE BY DESIGN

Raven does not replace review. It gives the team a faster, more reliable starting point by keeping every answer tied to the underlying source.

BUILT FOR

Shift Supervisor | Operator | Maintenance Planner | Safety Officer | Training Lead

PLANT QUESTION

What is the current temperature limit and protection context for A-102-B?

AI ANSWER | SOURCES | EQUIPMENT

ANSWER SUMMARY — 3 cited sources

- Current design limit for A-102-B at TI-1397 is 795 C. [1]
- TT/TAHH-1396 burner trip protects against coil outlet overtemperature. [2]
- FAL-L-1257A low-flow trip protects coils during low synthesis-gas flow. [3]
- Check older procedure references against the current design basis. [4]

REVIEW NOTE

Older procedure references should be checked against the current design basis before action.

Current revision verified

Current sources

Reformer Mechanical Design Basis.pdf Transition cone design temperature limits	Pg 11
A-102-B Reformer Operating Manual.pdf Coil outlet HH temperature protection	Pg 42
Ammonia Unit SOP - Start-up Heater.docx Low synthesis-gas flow trip logic	Pg 18

Equipment context

EQUIPMENT: A-102-B

AREA: Primary Reformer

KEY TAGS: TI-1397, TT/TAHH-1396, FAL-L-1257A

Safety impact

Decision speed

Teams get to the right source faster when the situation is time-sensitive and multiple references need to be checked.

Information reliability

Answers stay tied to the original document. When sources disagree, the discrepancy is visible and reviewable instead of being hidden in the workflow.

Knowledge continuity

Good answers no longer depend as heavily on which expert happens to be on shift, available, or still in the plant.

KPIS THIS IMPROVES

- **Time to cited answer** — faster access to the right source under time pressure
- **Document conflict detection** — inconsistencies surfaced before they become field decisions
- **Shift-independence** — answer quality less dependent on experience level or availability of specific people
- **Knowledge transfer readiness** — new and rotating staff work from the same plant context

PILOT SCOPE

See it on your plant's data

90-day pilot. One unit. Real documents. Measurable outcomes.

90-DAY PILOT

One unit, your documents. We load SOPs, P&IDs, vendor manuals, and recent records, then measure time to cited answer, citation accuracy, and escalation reduction on real plant questions.

CONTACT

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